JINGLIN ZHENG

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RESEARCH INTERESTS

AI Agents and Foundation Models | Trustworthy AI and Evaluation | Evidence-Grounded Decision-Making | Multimodal Agents

EDUCATION

University of Nottingham / University of Nottingham Ningbo China Sep 2023 - Jul 2027
BSc (Hons) Computer Science | 2+2 study pathway (UNNC Years 1-2; University of Nottingham, UK Years 3-4)

Selected completed marks: Algorithms, Data Structures and Efficiency (79); Fundamentals of Artificial Intelligence (79); Software Engineering Group Project (78); Introduction to Image Processing (74); Operating Systems & Concurrency (73); Mathematics for Computer Scientists (72).

Final-year coursework: Machine Learning; Designing Intelligent Agents; Individual Dissertation (40 credits).

RESEARCH EXPERIENCE

Independent Research - Reliable LLM Decision-Making Jul 2025 - Present
Initial research guidance from Prof. David Woodruff, Carnegie Mellon University

Three related projects spanning data-quality auditing, monitoring-signal evaluation, and calibration-data acquisition.

1. Context-Disjoint Answer-or-Abstain Evaluation | ADMA RDI

- Co-developed a data-quality audit of an archived 2,800-example QA pipeline, identifying context reuse and component exposure that compromised evaluation.
- Rebuilt context-disjoint evaluation with Train-only model selection, Validation-only thresholding, and a frozen 280-question Test set; evaluated an adapter-independent 13-feature monitor. **Short paper accepted at ADMA 2026 RDI.**

2. Generation-Derived Abstention Signals: Incremental Value and Cost | IDM

- Co-designed and conducted comparisons of direct scoring, 13 Base features, and 19 added dynamic features for Qwen3-8B and Llama-2-7B-Chat, using specialized factual/refusal DoRA adapters and matched sampling controls.
- Evaluated frozen monitors on 600 questions from 120 new RepLiQA documents; paired document-cluster intervals did not establish a full-dynamic ranking gain over cheaper inputs.
- Audited configuration and stopping sensitivity, unnecessary refusal, and runtime; identified a post-hoc reversal of the adapter comparison across joint prompt/serialization settings and measured 17-27x latency overhead for full monitors.

3. Calibration Budgets and Risk Certification | SAC

- Co-designed an equal-cost study crossing three document-breadth/question-depth allocations with KL, Hoeffding-Bentkus, and predictable-betting certifiers for two fixed LLM monitors.
- Conducted confirmation on 5,600 new documents across RepLiQA and DuoRC (28,000 questions; 288 experimental arms), testing how certifier choice changes the value of calibration-data allocation.
- Quantified allocation-certifier coverage interactions of -17.58 and -25.66 percentage points; assessed stability through resampling, mechanism controls, acquisition-cost analysis, and service-loss comparisons.

PUBLICATIONS & SUBMITTED MANUSCRIPTS

Accepted

Jinglin Zheng* and Liang Liao*. *Context-Disjoint Evaluation of LLM Answer-or-Abstain Routing: A Data-Quality Audit.* ADMA 2026, RDI Special Session. Accepted short paper.

Liang Liao* and **Jinglin Zheng***. *Auditing Generation-Derived Signals for LLM Abstention: Incremental Value and Cost.* IEEE Big Data 2026, 12th Special Session on Intelligent Data Mining (IDM). **Accepted regular paper (oral presentation).**

Submitted - decision pending

Calibration Budgets for LLM Answerability: Certifier Choice Changes Breadth-Depth Tradeoffs. Submitted to ACM SAC 2027. Decision pending.

* Equal contribution and shared first authorship; both authors are corresponding authors (ADMA and IDM).

SELECTED PROJECT

DR BEANS - AI-Integrated Inventory Management System Sep 2025 - May 2026

Software Engineering Group Project | Mark: 78/100 | Eight-member team

- Served as a core backend developer, designing backend functionality and interfaces that connected the inventory-management pipeline with an AI-agent component.
- Implemented Python-based backend services and integrated a local language model for conversational inventory queries and automated inventory notifications.
- Contributed to system architecture, integration, testing, and debugging, helping deliver a maintainable end-to-end system.

TECHNICAL SKILLS

Programming: Python, Java, C, Haskell, SQL

Machine Learning / LLMs: PyTorch, Hugging Face Transformers, PEFT (DoRA), scikit-learn, XGBoost, Sentence-Transformers; LLM fine-tuning, inference, and evaluation

Research Methods: Grouped cross-validation, bootstrap confidence intervals, ablations, risk-coverage/AURC analysis, uncertainty estimation, self-consistency, cost-sensitive evaluation, document-cluster risk certification, calibration-budget analysis

Software Development: Git, Flask, RESTful APIs, JUnit, backend development, system integration

HONORS & AWARDS

Excellent Freshmen Scholarship - Second Prize 2024-25 & 2025-26

University of Nottingham Ningbo China - awarded for two consecutive academic years in recognition of outstanding academic achievement.

LANGUAGE PROFICIENCY

IELTS Academic: 7.5 overall (Listening 8.0, Reading 8.0, Writing 6.5, Speaking 6.5)

STANDARDIZED TESTS

GRE General Test: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0

Test-centre administration at APCU-8115, Birmingham, United Kingdom (6 October 2026)